Software Engineering: Principles & Lecture Notes

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1. Course Outcomes & Syllabus Mapping (25CS312)

This reference compendium is structured directly around the 5 Course Outcomes of 25CS312:

" CO1: Choose relevant software process models in software development (Agile vs Waterfall).

" CO2: Build efficient software designs using various UML diagrams (Class, Sequence, Architecture).

" CO3: Build relevant testing strategies for verification and validation (Black-Box & White-Box).

" CO4: Apply quality assurance practices for robust software development (SQA, TPR, DRE).

" CO5: Evaluate software quality using relevant project estimation techniques (COCOMO II).

2. Learning Unit I: Software Process Models

Process models establish a systematic framework for managing software lifecycle development:

1. Linear Sequential (Waterfall): High rigidity; requirements frozen upfront; risk of late failure.
2. Incremental Model: Breaks product into functional increments delivered across timeboxed cycles.
3. Agile Process (Scrum / Kanban): Embraces emergent change, daily syncs, sprint planning, and empirical feedback. Selected by BODDU SOLOMON (URK25CS1191) for this project.

3. Architectural Design Patterns (LU II / CO2)

Modern web applications implement 4 distinct tiers to guarantee high cohesion and low coupling:

" Presentation Tier: React 18 with TypeScript and Tailwind CSS for responsive styling.

" State Tier: Zustand client store with persistent session caching and optimistic UI updates.

" Backend BaaS: Supabase PostgreSQL database protected by strict Row Level Security (RLS).

" Serverless AI: Deno Edge Functions with failover: Gemini 1.5 Flash -> OpenRouter -> NVIDIA NIM.